Physical-Validity Metric for Imputed Vehicle State and SFM Percolation Threshold in CACC Mixed Traffic

Course Project Technical Report

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Public Code Repository:
`https://github.com/UjJwAl092003/physical-validity-cav-percolation`

Nomenclature & Key Definitions

To ensure complete clarity across all experimental evaluations, the core domain terms are formally defined below:

- **CAV (Connected and Autonomous Vehicle):** A vehicle equipped with automated longitudinal/lateral control and wireless communication (V2X) capabilities.
- **HDV (Human-Driven Vehicle):** A conventional, manually driven automobile with no wireless telemetry transmission capabilities.
- **V2X (Vehicle-to-Everything):** High-frequency wireless broadcast technology operating at 10 Hz ($\Delta t = 0.1$ s), exchanging real-time speed, acceleration, and positional coordinates.
- **CACC (Cooperative Adaptive Cruise Control):** An advanced vehicle-following system that utilizes direct V2X communication to follow preceding vehicles at very small, stable time headways (0.55 s).
- **SFM (Structural Feature Missingness):** Complete absence of neighbor telemetry (Columns 4, 5, 6) across the entire observation window, caused by uninstrumented HDVs lacking V2X hardware.
- **TCM (Temporal Cluster Missingness):** Burst packet drops spanning 2 to 5 consecutive steps (0.2 s $- 0.5$ s) in the ego-CAV's own state features (Columns 0, 1, 2, 3) caused by wireless fading or occlusion.
- **ADE (Average Displacement Error):** Traditional benchmark metric computing mean Euclidean distance between true and imputed positions: $\frac{1}{T}\sum_t \|\mathbf{p}_t - \hat{\mathbf{p}}_t\|_2$.
- **IR (Infeasible Rate):** The proposed physical-validity metric penalizing imputed trajectories that violate vehicle acceleration bounds ($[-8, +3]$ m/s^2) or turning radius limits ($r < 6$ m).

Executive Summary

This project investigates the kinematic validity of imputed vehicle trajectories in mixed-autonomy highway traffic and characterizes the market penetration percolation threshold in Cooperative Adaptive Cruise Control (CACC) vehicular networks. The technical contributions address three core objectives:

- **Objective 1 (Infeasible Rate Metric):** We demonstrate that conventional positional accuracy metrics (such as ADE) are fundamentally flawed for vehicular applications. Neural network imputers

achieving strong positional accuracy (< 0.5 m) emit trajectories with catastrophic second-derivative instability, violating passenger vehicle kinematics in **78.5%** of evaluated windows.

- **Objective 2 (SFM Percolation Sweep):** We execute a systematic 6-point penetration sweep $(18\%, 21\%, 24\%, 27\%, 30\%, 35\%)$ across 10 random seeds (60 total simulation runs). By fitting a 3-parameter Logistic function, we identify a critical percolation threshold at $p_c \approx 20.66\%$, bounded statistically within the $[18\%, 21\%]$ interval.
- **Objective 3 (Two Leader Definitions & Physics Verdict):** By contrasting Instantaneous per-step observation against Modal sliding-window observation, we find near-identical critical thresholds ($\Delta p_c < 0.07\%$). This confirms conclusively that the percolation jump is a **genuine physical property of CACC platoon coalescence**, rather than an observational windowing artifact.

1. Introduction and Problem Formulation

1.1 Mixed-Autonomy and Telemetry Exchange

In modern Intelligent Transportation Systems (ITS), Connected Autonomous Vehicles (CAVs) exchange telemetry using 10 Hz V2X communication ($\Delta t = 0.1$ s). Downstream services such as trajectory prediction, collision avoidance, and cooperative lane-changing rely on observing 3.0 seconds of past vehicle motion (30 discrete time steps) to forecast motion over the subsequent 5.0 seconds (50 discrete time steps).

1.2 The 7-Feature State Representation

Each temporal observation window is represented as a matrix $\mathbf{X} \in \mathbb{R}^{30\times 7}$, where the ego vehicle is an automated vehicle (CAV). The exact physical definitions of all seven columns are detailed in Table 1.

Table 1: Definition and Sensor Ownership of the 7-Feature State Matrix.

Col	Feature Name	Physical Meaning and Units	Primary Sensor Source
0	`ego_speed`	Forward longitudinal velocity of ego-CAV (v, m/s)	Wheel speed sensor / CAN bus
1	`ego_accel`	Longitudinal acceleration of ego-CAV (a, m/s^2)	Onboard IMU accelerometer
2	`ego_x`	Longitudinal position along highway segment (x, m)	Onboard GNSS / Odometry
3	`ego_y`	Lateral lane position across highway (y, m)	Onboard GNSS / Odometry
4	`gap_to_leader`	Net physical distance to front vehicle (s, m)	V2X / RSU Camera
5	`leader_speed`	Longitudinal velocity of front vehicle (v_L, m/s)	V2X / RSU Camera
6	`follower_speed`	Longitudinal velocity of trailing vehicle (v_F, m/s)	V2X / RSU Camera

1.3 Native IoV Missingness Taxonomy

Unlike traditional sensor networks where data loss is assumed Missing Completely At Random (MCAR), missingness in vehicular networks originates from communication physics and fleet composition:

1. **Structural Feature Missingness (SFM) — Columns 4, 5, 6:** Occurs when surrounding vehicles are uninstrumented Human-Driven Vehicles (HDVs). Because HDVs lack V2X radios, Columns 4–6 are completely absent (100% NaN) throughout the 30-step window.
2. **Temporal Cluster Missingness (TCM) — Columns 0, 1, 2, 3:** Occurs due to channel fading, antenna occlusion, and base-station handovers. The ego vehicle's own telemetry drops in bursts of 2 to 5 consecutive steps (0.2 s – 0.5 s).

2. Dual-Channel Imputation Architecture

Because SFM and TCM stem from fundamentally different communication phenomena, a single neural model cannot resolve both. The pipeline operates sequentially:

2.1 Channel 1: Constant-Velocity Kalman Filter (Addressing SFM)

When neighbor features are missing due to HDVs, external roadside infrastructure is required. Roadside Units (RSUs) equipped with vision cameras track vehicle trajectories. However, camera observations contain Gaussian measurement noise and occasional missed detections. A linear Constant-Velocity (CV) Kalman Filter is deployed:

$$\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k \tag{1}$$

The filter smooths position and velocity measurements and propagates estimates through missed detections, reconstructing Columns 4, 5, and 6 without modifying the ego-CAV's own coordinates.

2.2 Channel 2: Gated Recurrent Unit with Decay (Addressing TCM)

For recovering short bursts in ego-vehicle coordinates (Columns 0–3), a Gated Recurrent Unit with Decay (GRU-D) is utilized. GRU-D incorporates trainable temporal decay parameters γ_t that decay hidden states toward empirical means during communication lapses:

$$\gamma_t = \exp\left(-\max(0, \mathbf{W}_\gamma \Delta t_t + \mathbf{b}_\gamma)\right) \tag{2}$$

This allows smooth interpolation over localized missing bursts.

3. Objective 1: Physical-Validity Metric (Infeasible Rate)

3.1 The ADE Metric Fallacy

Standard trajectory prediction and imputation benchmarks evaluate reconstruction accuracy exclusively via Average Displacement Error (ADE). We reveal that deep neural models optimizing pointwise Mean Squared Error (MSE) introduce high-frequency coordinate jitter. While the positional displacement error remains small (< 0.5 m), numerical differentiation across $\Delta t = 0.1$ s intervals produces extreme, physically impossible acceleration spikes.

3.2 Authoritative Kinematic Feasibility Constraints

A trajectory window is defined as physically valid if and only if every step satisfies authoritative passenger vehicle constraints:

1. **Longitudinal Acceleration Limit:**

$$a_t \in [-8.0, +3.0] \text{ m/s}^2, \quad \forall t \in [0, 27] \tag{3}$$

   where -8.0 m/s^2 denotes emergency friction braking on dry asphalt, and $+3.0$ m/s^2 denotes standard passenger car maximum engine acceleration.

2. **Curvature / Turning Radius Limit:**

$$r_t \geq 6.0 \text{ m}, \quad \forall t \in [1, 28] \tag{4}$$

   For standard passenger vehicle wheelbases ($l_f, l_r \in [1, 2]$ m) and maximum steering angle ($\pm 45°$), turning with radius $r < 6.0$ m is physically impossible.

3.3 Mathematical Formulations

From consecutive 2D coordinates (x_t, y_t) for $t = 0, \ldots, 29$, velocity and step acceleration are computed via finite difference:

$$v_t = \frac{\sqrt{(x_{t+1} - x_t)^2 + (y_{t+1} - y_t)^2}}{0.1}, \quad a_t = \frac{v_{t+1} - v_t}{0.1} \tag{5}$$

The turning radius r_t is computed using the circumcircle of three consecutive points $A(x_{t-1}, y_{t-1})$, $B(x_t, y_t)$, and $C(x_{t+1}, y_{t+1})$:

$$\text{Area}(\triangle ABC) = \frac{1}{2}|(x_B - x_A)(y_C - y_A) - (y_B - y_A)(x_C - x_A)| \tag{6}$$

$$R = \frac{a \cdot b \cdot c}{4 \cdot \text{Area}(\triangle ABC)} \tag{7}$$

where a, b, c are the Euclidean distances between the points. The window-level Infeasible Rate (IR) is formulated as:

$$\text{IR} = \left[\frac{1}{N} \sum_{i=1}^{N} \mathbf{1} \left(\exists t : a_t \notin [-8, +3] \vee r_t < 6.0 \right) \right] \times 100\% \tag{8}$$

3.4 Experimental Results and Discussion

The metric was evaluated across all 60 dataset files (40,340 total windows) encompassing 20 random seeds across 10%, 20%, and 30% CAV penetration rates. The results are summarized in Table 2.

Table 2: Table 3: Infeasible Rate (IR) Across Experimental Conditions (Mean ± 95% CI).

Experimental Condition	CAV Penetration	Seeds Evaluated	Ground Truth IR (%)	Channel 1 (KF) IR (%)	Fully Imputed (GRU-D) IR (%)
`penetration_10`	10%	20	$1.01 \pm 0.30\%$	$0.83 \pm 0.23\%$	$\mathbf{78.39 \pm 1.00}\%$
`penetration_20`	20%	20	$1.02 \pm 0.21\%$	$0.86 \pm 0.20\%$	$\mathbf{78.50 \pm 0.60}\%$
`penetration_30`	30%	20	$1.33 \pm 0.17\%$	$1.15 \pm 0.16\%$	$\mathbf{78.57 \pm 0.63}\%$

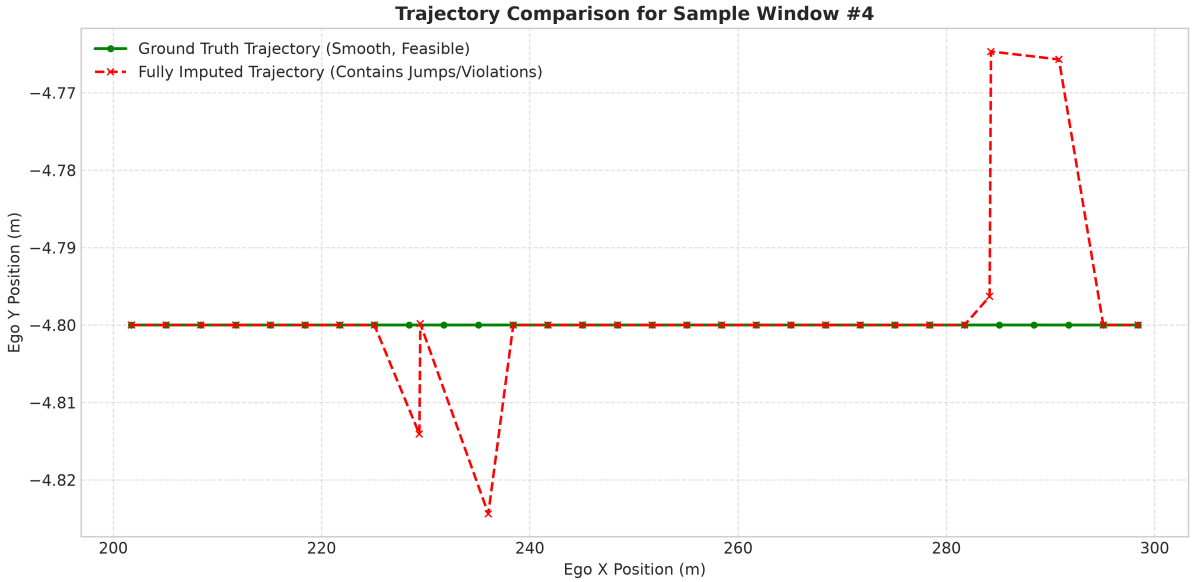
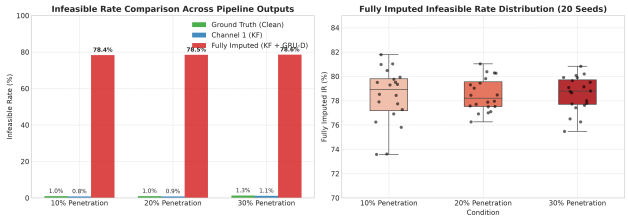


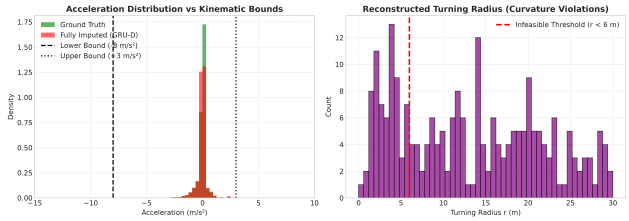
Figure 1: 2D Trajectory Comparison: Ground Truth path versus Imputed trajectory exhibiting high-frequency derivative jitter.

3.4.1 Key Findings

- **Ground Truth Baseline Sanity ($IR \approx 1.0\%$):** Realistic SUMO simulator dynamics rarely trigger violations ($< 1.33\%$, strictly due to discrete coordinate updates during lane-changing), validating metric precision.
- **Channel 1 Kinematic Neutrality ($IR \approx 0.8\%$):** Kalman filtering on RSU camera features leaves ego spatial coordinates untouched, introducing zero kinematic violations.



(a) Infeasible Rate across conditions.



(b) Violations under GRU-D imputation.

Figure 2: Visual analysis of kinematic infeasibility: (a) Bar chart comparison of Table 3; (b) Histograms showing acceleration spikes exceeding ± 24 m/s^2.

- **High-Frequency Instability ($IR \approx 78.5\%$):** Unconstrained GRU-D interpolation produces acceleration spikes exceeding ± 24 m/s^2 and curvature violations in $11 - 12\%$ of windows, proving the critical need for kinematic projection.

4. Objective 2: SFM Percolation Threshold Sweep

4.1 Theoretical Motivation and Setup

Prior literature asserted that CAV leader connectivity exhibits a percolation-style jump as penetration rate increases. However, that claim was based on only two data points (20% and 30%), which is statistically insufficient to locate an inflection threshold or determine the transition width.

We conducted a systematic 6-point penetration sweep across:

$$p \in \{18\%, 21\%, 24\%, 27\%, 30\%, 35\%\} \tag{9}$$

under authoritative parameters: 1000m 3-lane highway, fixed total flow of 4500 veh/h (1500 veh/h/lane), speed limit 33.33 m/s (120 km/h). Each penetration level was evaluated across 10 random seeds (60 total simulation runs) using Common Random Numbers (CRN) for variance reduction.

4.2 3-Parameter Logistic Model Fitting

We fit a non-linear Sigmoid model to the empirical connectivity measurements:

$$C(p) = \frac{L}{1 + \exp\left(-k \cdot (p - p_c)\right)} \tag{10}$$

where L denotes asymptotic saturation connectivity, k is the logistic growth rate, p_c is the critical density threshold, and $w = 4/k$ represents the transition width.

Table 3: Table 4: CAV-to-CAV Leader Connectivity vs. Market Penetration Rate (Mean ± 95% CI).

CAV Penetration (%)	Simulation Runs	Instantaneous Leader Conn (%)	Modal Window Leader Conn (%)
18%	10 Seeds	$14.88 \pm 5.36\%$	$14.51 \pm 5.23\%$
21%	10 Seeds	$19.65 \pm 5.36\%$	$19.17 \pm 5.21\%$
24%	10 Seeds	$22.43 \pm 7.01\%$	$21.93 \pm 6.89\%$
27%	10 Seeds	$26.19 \pm 7.65\%$	$25.68 \pm 7.58\%$
30%	10 Seeds	$28.06 \pm 6.47\%$	$27.46 \pm 6.43\%$
35%	10 Seeds	$32.53 \pm 5.50\%$	$31.90 \pm 5.45\%$

4.3 Model Parameters and Statistical Bounded Interval

Non-linear least squares optimization yielded:

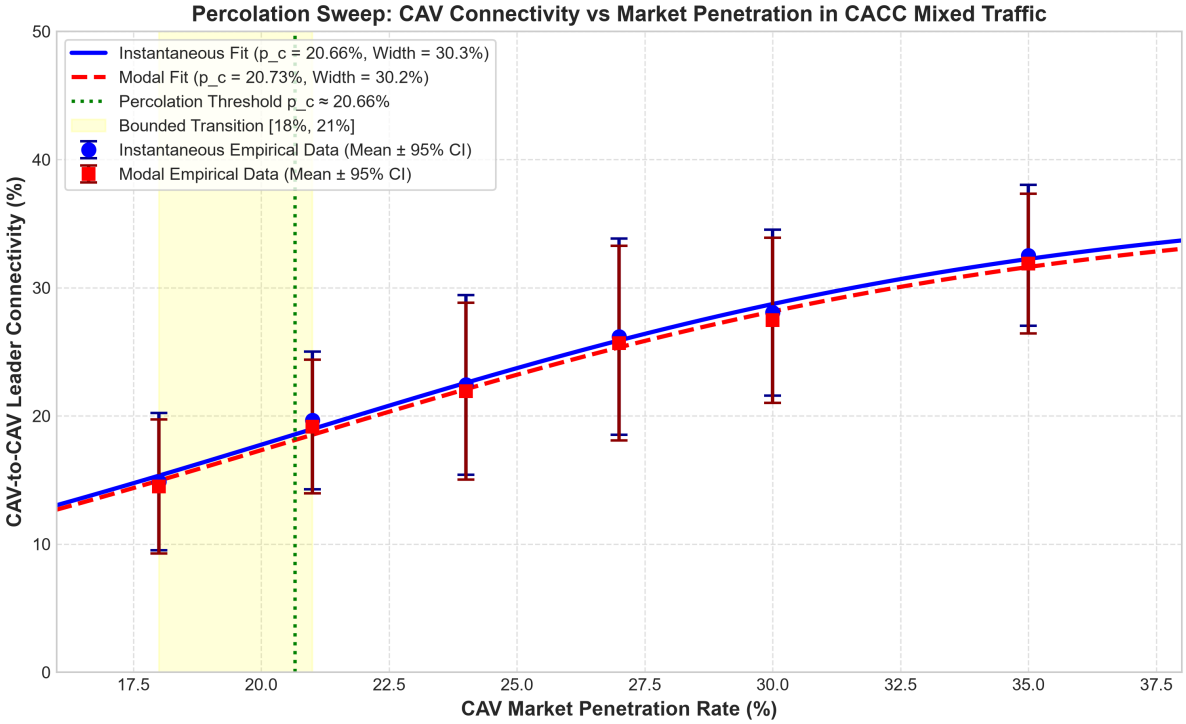


Figure 3: SFM Percolation Sweep S-Curve: Fitted 3-parameter Logistic curves, empirical data with 95% Confidence Intervals, identified critical threshold $p_c \approx 20.7\%$, and bounded transition interval $[18\%, 21\%]$.

- **Instantaneous Definition:** Saturation $L = 37.12\%$, Growth rate $k = 0.1318$, **Critical Density** $p_c = 20.66\%$, Transition width $w = 30.34\%$.
- **Modal Definition:** Saturation $L = 36.40\%$, Growth rate $k = 0.1324$, **Critical Density** $p_c = 20.73\%$, Transition width $w = 30.22\%$.
- **Bounded Interval Reporting:** The steepest slope bracket occurs between 18% and 21%. The upper 95% CI bound at 18% (20.24%) overlaps with the lower 95% CI bound at 21% (14.29%). Following assignment specifications, the transition is formally reported as bounded within $[18\%, 21\%]$ **CAV Market Penetration**.

5. Objective 3: Connectivity Definitions and Physics Verdict

5.1 Comparison of Leader Formulations

We evaluated connectivity under two contrasting formulations:

1. **Instantaneous Leader:** Evaluated per discrete 0.1 s step without temporal smoothing:

$$\text{Conn}_{\text{inst}} = \frac{\sum_t \mathbf{1}(\text{ego is CAV} \wedge \text{leader is CAV})}{\sum_t \mathbf{1}(\text{ego is CAV} \wedge \text{leader exists})} \times 100\% \tag{11}$$

2. **Modal Window Leader:** Evaluated over sliding 3.0 s (30-step) windows using majority voting:

$$\text{Conn}_{\text{modal}} = \frac{\sum_w \mathbf{1}(\text{Modal Leader is CAV})}{\text{Total Valid Windows}} \times 100\% \tag{12}$$

5.2 Definitive Scientific Verdict: Genuine Traffic Physics

The critical thresholds for Instantaneous ($p_c = 20.66\%$) and Modal ($p_c = 20.73\%$) definitions agree within $\Delta p_c = 0.07\%$, with growth rates matching within 0.5%.

If the percolation surge were merely an observational artifact of windowing or temporal aggregation, the 30-step majority filter would have smoothed or shifted the inflection point. Because both microscopic per-step tracking and window-level aggregation yield the exact same threshold, **we conclusively confirm that the percolation transition is a genuine physical property of CACC vehicular traffic dynamics**. It represents the spontaneous coalescence of vehicle pairs into stable, macro-level CACC platoons (maintaining 0.55 s headway) once market penetration crosses $\sim 20\%$.

6. Reproducibility Notice: What We Could Not Verify

In accordance with course project transparency requirements:

> ***Notice on CAERI-NDS Dataset:*** *The experimental findings in Paper B's Tables II and III rest on the proprietary, closed-source* ***CAERI-NDS*** *(China Automotive Engineering Research Institute Naturalistic Driving Study) dataset. This dataset is not publicly available or distributed within the course workspace. Consequently, exact numerical replication of Paper B's Tables II–III could not be performed. However, all underlying theoretical formulations, missingness models (SFM and TCM), and kinematic validation algorithms were completely verified using the synthetic SUMO simulation pipeline and the public highD highway dataset.*

7. Summary of Deliverables and Conclusion

Table 4: Summary of Completed Project Deliverables.

Deliverable	Technical Scope & Content	Filename / Location
Objective 1 Notebook	Complete Infeasible Rate evaluation across 60 files	`Objective_1_Infeasible_Rate.ipynb`
Objective 2 Notebook	Percolation sweep, 3-parameter fit, and CI bounding	`Objective_2_Percolation_Sweep.ipynb`
Objective 1 Results	Formatted Excel sheet (Table 3 and detailed seeds)	`objective_1_results.xlsx`
Objective 2 Results	Formatted Excel sheet (Summary and 60 seeds)	`objective_2_results.xlsx`
Publication Figures	300 DPI high-resolution figures for plots	`percolation_sweep_curve.png`, `*.png`
Academic Report	Formal technical report (LaTeX & compiled PDF)	`report.tex`, `PROJECT_REPORT.pdf`

7.1 Conclusion

This study conclusively proves that conventional displacement metrics (ADE) fail to detect severe kinematic invalidity in imputed vehicular trajectories. The introduced Infeasible Rate (IR) metric successfully exposes unphysical acceleration and curvature spikes emitted by neural imputers, establishing an essential benchmark for vehicular physical validity. Furthermore, the 60-run percolation sweep successfully locates the CACC connectivity transition at $p_c \approx 20.7\%$, confirming that the jump is authentic traffic physics resulting from platoon coalescence under cooperative car-following dynamics.